Exercise 1

Variables, Collections & Conditionals

Data Analytics with Python · Databricks Environment

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Instructions

Work through each section in your Databricks notebook. Create a new cell for each task. Read each question carefully before writing your code.

- You may refer to the notes and examples shown in class.
- Use print() to display your results where asked.
- Write your answers (expected output) in the spaces provided below.
- Complete this document using Microsoft word or google documents.
- Save the document as PDF
- Upload both the PDF and databricks notebook to GitHub
- Send GitHub link, name, course name to (mvuko@brightlearn.co.za)

The Dataset

Below is a small student dataset. You will manually enter this data as Python variables and collections in your notebook. Each row represents one student's record.

student_name	age	subject	score	passed
"Amara"	19	"Math"	72	True
"Ben"	21	"Science"	55	False
"Chloe"	20	"Math"	88	True
"David"	22	"English"	45	False
"Eva"	19	"Science"	91	True
"Farai"	23	"English"	63	True
"Grace"	20	"Math"	50	False
"Hassan"	21	"Science"	77	True

Note: *This is the only data you will work with in this exercise.*

Section 1 — Variables & Data Types

In this section, you will create individual variables for a single student's record and explore Python data types.

Task 1.1 — Create variables for Amara's record

In a new cell, create the following variables exactly as shown:

```
student_name = "Amara"
age          = 19
subject      = "Math"
score        = 72
passed       = True
```

Task 1.1 After running the cell, what is the data type of each variable? Write it next to each variable name.

a. student_name → ___________
b. age → ___________
c. score → ___________
d. passed → ___________

Your answer:

a. student_name → <class 'str'>
b. age → <class 'int'>
c. score → <class 'int'>
d. passed → <class 'bool'>

Task 1.2 — Use type() and print()

Add a new cell and use the type() function to confirm the data type of score and passed. Then use print() to display the result.

Task 1.2 Write the code you used and record the output below.

e. Code: print(type(score)) and print(type(passed))
f. Output for type(score): <class 'int'>
g. Output for type(passed): <class 'bool'>

Task 1.3 — Simple arithmetic

A student's final mark is calculated as their score divided by 100, multiplied by 150 (the exam is out of 150).

Task 1.3 Write a single line of Python that calculates Amara's final mark and stores it in a variable called final_mark. Then print it.

h. What is Amara's final_mark? ___108.0________
i. What data type is final_mark? _____<class 'float'>______

Section 2 — Collections

In this section you will store the dataset in Python lists and a dictionary, and access specific values.

Task 2.1 — Create a list of student names

Create a list called student_names that contains all 8 student names from the dataset, in order.

Task 2.1 Answer the following questions about your list:

j. What is student_names[0]? ___________
k. What is student_names[-1]? ___________
l. What is student_names[2:5]? ___________
m. What is len(student_names)? ___________

Your answer:

student_names = ["Amara", "Ben", "Chloe", "David", "Eva", "Farai", "Grace", "Hassan"]

Amara

Hassan

['Chloe', 'David', 'Eva']

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Task 2.2 — Create a list of scores

Create a list called scores containing all 8 scores in the same order as the names list.

Task 2.2 Using your scores list, answer:

n. What is scores[0]? ___________

o. What is scores[4]? ___________

p. What is the result of scores[0] + scores[4]? ___________

Your answer:

scores = [72, 55, 88, 45, 91, 63, 50, 77]

print(scores[0])

print(scores[4])

print((scores[0]) + (scores[4]))

Task 2.3 — Create a dictionary for one student

Create a dictionary called hassan that stores all five fields for Hassan as key-value pairs.

```
hassan = {
    "student_name": ...,
    "age": ...,
    "subject": ...,
    "score": ...,
    "passed": ...
}
```

Task 2.3 Using your dictionary, answer:

q. What does hassan["subject"] return? ___________

r. What does hassan["score"] + 10 return? ___________

s. What does type(hassan["passed"]) return? ___________

Your answer:

Science

87

<class 'bool'>

Task 2.4 — A list of dictionaries

Create a list called students where each element is a dictionary representing one student. Include only Amara, Ben, Chloe, and David.

Task 2.4 Using your list of dictionaries, answer:

t. What is students[1]["score"]? ___________

u. What is students[0]["student_name"]? ___________

v. How many items are in students? ___________

Your answer:

55

Amara

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Section 3 — Logical Operators & Conditionals

In this section you will use comparison operators, logical operators (and, or, not), and if/elif/else statements to analyse student records.

Task 3.1 — Comparison expressions

Using the variables you created in Section 1 (Amara's record), evaluate the following expressions. Write the result (True or False) without running the code first — then verify in your notebook.

Task 3.1 Predict the output of each expression:

score > 60 → ____True______

age == 20 → ___False________

subject != "Science" → ___False________

score >= 72 → ____True________

Your answer:

(Verified from Notebook)

True

False

True

True

Task 3.2 — Logical operators

Still using Amara's variables, evaluate these combined expressions:

Task 3.2 Predict and verify:

w. score > 60 and passed == True → ___________

x. age < 18 or score > 70 → ___________

y. not passed → ___________

z. score > 80 and subject == "Math" → ___________

Your answer:

True

True

False

False

Task 3.3 — Write an if/elif/else statement

Write a conditional statement that prints a grade category based on a student's score. Use the following rules:

Score Range	Grade
75 and above	**Distinction**
50 – 74	**Pass**
Below 50	**Fail**

Test your conditional using score = 72 (Amara) and again with score = 45 (David).

Task 3.3 Write your if/elif/else code below and record the output for each test:

aa. Output when score = 72: ___________

bb. Output when score = 45: ___________

cc. Output when score = 91: ___________

Your answer:

Pass

Fail

Distinction

Task 3.4 — Challenge: combined condition

Write a conditional statement that checks two conditions at once: the student scored above 60 AND their subject is "Math". If both are true, print "Math high achiever". Otherwise print "Does not meet criteria".

Task 3.4 Test with Amara (score=72, subject="Math") and Ben (score=55, subject="Science"):

dd. Output for Amara: ___________

ee. Output for Ben: ___________

ff. What logical operator did you use? ___________

Your answer:

Math high achiver

Does not meet criteria

Where score = 72 and 55

```
if score > 60 and subject == "Math":
    print("Math high achiver")
else:
    print("Does not meet criteria")
```

Reflection

Answer these short questions in your own words — no code needed.

Task R1 What is the difference between a list and a dictionary in Python? Give one example of when you would choose each.

Your answer:

List: Ordered, usually identified by [] bracket, can be accessed by index positin [0,1…] and used where order is required such as people's names.

Dictionary: they are used when labelling data is required, so they are written in pairs such as {"name": "Amara"} where the name=Key and Amara=Value, and they also use or identified by {} brackets.

Task R2 In Task 3.2, explain in plain English what this expression checks: score > 60 and passed == True

Your answer:

It checks both conditions at the same time. This means if the score is greater than 60 AND passed, both “greater than” 60 AND “passed” (are equally true) == True

Task R3 If you wanted to store all 8 students and be able to look up any student by name quickly, which collection type would you use? Why?

Your answer:

Dictionaries, because it allows you to look up for names using a key, which would have stored all of those names.

Good luck!